

Kabadiwala Connect

7–10 Minute Problem & Solution Pitch

Smart India Hackathon 2026 • Problem Statement 26229 • Team Raddi Rangers

Purpose: This round is problem- and solution-focused. The goal is to make the judges clearly understand the real-world gap, what field research revealed, and why Kabadiwala Connect is the right solution.

Core message:

“The problem isn't that the recycling ecosystem doesn't exist. The problem is that its layers aren't digitally connected.”

1. Opening — The Real Problem (0:00–0:45)

Say:

“Good morning respected judges.

When we talk about e-waste recycling, we usually think about the recycler at the end of the chain. But we asked ourselves a different question: **Where does that e-waste actually begin its journey?**

It often begins with a door-to-door kabadiwala. They collect small quantities of electronic scrap from households and sell it onward to a local dealer. The dealer aggregates it, and eventually some of that material reaches an authorized recycler.

So the ecosystem already exists. **The problem is that the digital connection between these layers is missing.**

And that missing connection creates problems with price visibility, aggregation, traceability and formal handover.”

2. Field Research — We Went to the Ground (0:45–1:30)

Transition:

“Instead of assuming what the problem was, we went to the field to understand how this chain actually works.”

■ **PLAY THE 30–40 SECOND FIELD-RESEARCH VIDEO HERE.**

After the video:

“What you just saw helped us understand something important: the informal ecosystem isn't an isolated problem waiting to be replaced. It is an existing network of collectors, dealers and buyers. The opportunity is to connect that network to the formal recycling system.”

3. What We Discovered (1:30–2:30)

Say:

“The first thing we understood is that the individual kabadiwala is not the right aggregation point for formal recycling. They collect small quantities from many households.

A dealer, on the other hand, aggregates material from multiple collectors and therefore has the volume required to approach recyclers.

But there are gaps at both ends.

At the collector level: there is limited access to reliable price information and simple safety guidance.

At the dealer level: transactions and stock can be difficult to organize, especially when connectivity is unreliable.

At the recycler level: there is a need for structured lots and reliable information about what is actually being handed over.

So we realized that this is not three separate problems. It is one broken chain."

4. Define the Problem Clearly (2:30–3:00)

Say:

"Our problem is the lack of a simple digital bridge that connects the informal first-mile collector, the dealer who aggregates the material, and the authorized recycler who formally processes it.

We are not trying to replace the existing informal ecosystem.

We are trying to formalize the transaction that is already happening."

5. Our Solution — Kabadiwala Connect (3:00–4:30)

Say:

"That is where Kabadiwala Connect comes in.

We designed it as a two-level solution because our research showed us that one application cannot treat every participant in the same way.

For the individual kabadiwala: we keep it extremely simple. They can take a photograph of the scrap. Our on-device AI suggests the material category. But AI is only an assistant—the collector can confirm or correct the suggestion. If the model isn't confident, we fall back to manual selection.

We also provide simple price and safety information in a vernacular interface, so the collector can make a more informed decision.

For the dealer: the dealer records purchases from collectors. Those purchases become stock. Once enough material is available, the dealer creates a lot. The system can then help identify suitable recyclers based on factors such as accepted material, rate, distance and pickup availability."

6. The Dealer Is the Bridge (4:30–5:30)

Say:

"And this is actually the key to our solution.

We are not trying to make every individual kabadiwala directly deal with a recycler.

Instead, the dealer becomes the aggregation bridge.

Small quantities from multiple collectors are recorded and consolidated into a recycler-ready lot. This makes the existing ecosystem more compatible with the way formal recyclers actually operate.”

Visual: Small quantities from many collectors → Dealer → Aggregated lot → Authorized recycler.

7. Digital Handover & Traceability (5:30–6:15)

Say:

“Once the dealer selects a recycler, the lot gets a unique digital identity through a QR code. The QR contains only the Lot ID.

At the physical handover, the recycler scans it and retrieves the lot information. The dealer's declared weight and the recycler's physically verified weight are recorded separately.

If the discrepancy is more than 30 percent, the system raises a warning rather than silently accepting the difference.

Once the recycler confirms the handover, we have a structured digital record of that transaction.

So we are not just helping someone sell scrap. We are creating continuity of information across the recycling chain.”

8. Why Offline-First? (6:15–6:50)

Say:

“There is another important insight from our target environment: connectivity cannot be assumed everywhere.

So the dealer workflow is designed to be offline-first. A purchase can be recorded locally, and once connectivity returns, it can synchronize with the central system.

This means the solution adapts to the ecosystem instead of expecting the ecosystem to adapt to perfect internet connectivity.”

9. What Makes Us Different? (6:50–7:30)

Say:

“Existing digital platforms can help with buying, selling or price discovery. But our focus is different.

We are focusing on the missing link between the informal collection layer and formal recycling.

First, we start at the informal layer—the person who is currently least visible digitally.

Second, we use the dealer as the aggregation bridge instead of forcing small collectors to behave like large recyclers.

And third, we create traceability across the journey:

Collector → Purchase → Stock → Lot → Recycler → Verified Handover."

10. Impact (7:30–8:15)

Say:

"The impact is not limited to one user.

For the collector: better price visibility and accessible safety information.

For the dealer: better organization of purchases, stock and bulk lots.

For the recycler: structured incoming lots and documented handovers.

For the ecosystem: a stronger bridge between informal collection and authorized recycling.

Ultimately, better traceability means better accountability for where material goes."

11. Conclusion (8:15–8:45)

Say:

"So when we started this problem, we didn't ask: 'How can we build another recycling app?'

We asked: **'Why does the person who first collects the e-waste have so little digital connection to the person who formally recycles it?'**

Our answer is Kabadiwala Connect.

We don't replace the existing chain. We connect it.

From the first collection, to the dealer, to aggregation, to the recycler, and finally to a traceable digital handover.

That is the bridge we want to build.

Thank you."

Presentation Notes

- Keep this round problem- and solution-focused; do not turn it into a detailed prototype walkthrough.
- Use the field-research video as evidence that the problem was investigated rather than assumed.
- Explain AI as an assisting tool for the collector; the collector can confirm/correct it.
- Emphasize the dealer as the aggregation bridge.
- Do not spend time on database schemas, APIs, UUIDs or Git unless judges ask.
- Only claim features as implemented if they are actually implemented. Describe unimplemented submitted features as proposed/planned.
- If asked about EPR, distinguish a platform-verified transaction/traceability record from an official government-issued EPR certificate unless actual official integration exists.

One-Line Takeaway

“The recycling ecosystem already exists; Kabadiwala Connect builds the digital bridge that connects its informal collection layer to formal recycling.”